Genessa 250mg Film Coated Tablets

1. NAME OF THE MEDICINAL PRODUCT

Genessa 250mg Film Coated Tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each tablet contains 250mg of Gefitinib.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Film coated tablet.

Round, biconvex, brown film-coated tablet debossed with "GEF 250" on one side and plain on other side.

4. CLINICAL PARTICULARS

4.1 Therapeutic Indications

Genessa is indicated for the treatment of adult patients with locally advanced or metastatic non-small cell lung cancer (NSCLC) with activating mutations of EGFR-TK.

4.2 Posology and Method of Administration

The recommended dose of Genessa is one 250 mg tablet once a day, taken with or without food. If a dose of Genessa is missed, it should be taken as soon as the patient remembers. If it is less than 12 hours to the next dose, the patient should not take the missed dose. Patients should not take a double dose (two doses at the same time) to make up for a forgotten dose.

Where dosing of whole tablets is not possible, such as patients who are only able to swallow liquids, tablets may be administered as a dispersion in water. The tablet should be dropped into half a glass of drinking water (non-carbonated), without crushing, and the glass stirred until the tablet has dispersed (approximately 15 minutes) and the contents subsequently drunk immediately. The glass should be rinsed with a further half glass of water and the contents drunk. The liquid can also be administered via a nasogastric tube.

Genessa is not recommended for use in children or adolescents as safety and effectiveness in these patient populations has not been studied.

No dosage adjustment is required on the basis of patient age, body weight, gender, ethnicity or renal function or in patients with moderate to severe hepatic impairment due to liver metastases.

Dosage adjustment: Patients with poorly tolerated diarrhea or skin adverse drug reactions may be successfully managed by providing a brief (up to 14 days) therapy interruption followed by reinstatement of the 250 mg dose.

4.3 Contraindications

Known severe hypersensitivity to the active substance or to any of the excipients of this product.

4.4 Special Warnings and Precautions for Use

When considering the use of Genessa as first-line treatment for locally advanced or metastatic NSCLC, it is recommended that EGFR mutation assessment of the tumor tissue is attempted for all patients. When assessing the mutation status of a patient it is important that a well-validated and robust methodology is chosen to minimize the possibility of false negative or false positive determinations. Tumor samples which are used for the diagnosis of advanced NSCLC are the preferred sample type for EGFR mutation testing. A tumor sample should be collected and tested where possible. If a tumor sample is not available or evaluable, then circulating tumor DNA (ctDNA) obtained from a blood (plasma) sample may be used. Only robust, reliable, sensitive test(s) with demonstrated utility on ctDNA should be used for the determination of EGFR mutation status of ctDNA. EGFR mutations identified in ctDNA are highly predictive of EGFR mutation positive tumors. However, it is not always possible to detect EGFR mutations using this sample type (0.2% false positives, 34.3% false negatives).

Interstitial Lung Disease (ILD), which may be acute in onset, has been observed in patients receiving Gefitinib, and some cases have been fatal. If patients present with worsening of respiratory symptoms such as dyspnea, cough and fever, Genessa should be interrupted and prompt investigation initiated. If ILD is confirmed, Genessa should be discontinued and the patient treated appropriately.

In a study in patients with NSCLC who were followed up for 12 weeks when receiving Gefitinib or chemotherapy, the following risk factors for developing ILD (irrespective of whether the patient received Gefitinib or chemotherapy) were identified: smoking, poor performance status (PS $\geq$ 2), CT scan evidence of reduced normal lung ($\leq$ 50%), recent diagnosis of NSCLC (< 6 months), pre-existing ILD, increasing age ($\geq$ 55 years old) and concurrent cardiac disease. Risk of mortality among patients who developed ILD on both treatments was higher in patients with the following risk factors: smoking, CT scan evidence of reduced normal lung ($\leq$ 50%), pre-existing ILD, increasing age ($\geq$ 65 years old), and extensive areas adherent to pleura ($\geq$ 50%).

Liver function test abnormalities (including increase in alanine aminotransferase, aspartate aminotransferase, bilirubin) have been observed, uncommonly presenting as hepatitis. There have been isolated reports of hepatic failure which in some cases led to fatal outcomes. Therefore, periodic liver function testing is recommended. Genessa should be used cautiously in the presence of mild to moderate change in liver function. Discontinuation should be considered if changes are severe.

Cerebrovascular events have been reported in clinical studies of Gefitinib. A relationship with Gefitinib has not been established.

Substances that are inducers of CYP3A4 activity may increase metabolism and decrease Gefitinib plasma concentrations. Therefore, co-medication with CYP3A4 inducers (e.g.

phenytoin, carbamazepine, rifampicin, barbiturates or St John's Wort) may reduce efficacy.

International Normalized Ratio (INR) elevations and/or bleeding events have been reported in some patients taking warfarin. Patients taking warfarin should be monitored regularly for changes in Prothrombin Time (PT) or INR.

Drugs that cause significant sustained elevation in gastric pH may reduce plasma concentrations of Gefitinib and therefore may reduce efficacy.

Patients should be advised to seek medical advice promptly in the event of developing:

- severe or persistent diarrhea, nausea, vomiting or anorexia

These symptoms should be managed as clinically indicated.

Patients presenting with signs and symptoms suggestive of keratitis such as acute or worsening: eye inflammation, lacrimation, light sensitivity, blurred vision, eye pain and/or red eye should be referred promptly to an ophthalmology specialist.

If a diagnosis of ulcerative keratitis is confirmed, treatment with Genessa should be interrupted, and if symptoms do not resolve, or recur on reintroduction of Genessa, permanent discontinuation should be considered.

In a study of Gefitinib and radiation in pediatric patients, newly diagnosed with brain stem glioma or incompletely resected supratentorial malignant glioma, 4 cases (1 fatal) of CNS hemorrhages were reported from 45 patients enrolled. A further case of CNS hemorrhage has been reported in a child with an ependymoma from a trial with Gefitinib alone. An increased risk of cerebral hemorrhage in adult patients with NSCLC receiving Gefitinib has not been established.

In a study where Gefitinib and vinorelbine have been used concomitantly, indicate that Gefitinib may exacerbate the neutropenic effect of vinorelbine.

Gastrointestinal perforation has been reported in patients taking Gefitinib. In most cases this is associated with other known risk factors, including increasing age, concomitant medications such as steroids or NSAIDS, underlying history of GI ulceration, smoking or bowel metastases at sites of perforation.

4.5 Interaction with Other Medicinal Products and Other Forms of Interaction

In vitro studies have shown that the metabolism of gefitinib is predominantly via CYP3A4.

Co-administration with rifampicin (a known potent CYP3A4 inducer) in healthy volunteers reduced mean gefitinib AUC by 83% of that without rifampicin.

Co-administration with itraconazole (a CYP3A4 inhibitor) resulted in an 80 % increase in the mean AUC of gefitinib in healthy volunteers. This increase may be clinically relevant since adverse experiences are related to dose and exposure.

Co-administration of ranitidine at a dose that caused sustained elevations in gastric pH (≥ 5), resulted in a reduced mean gefitinib AUC by 47% in healthy volunteers.

INR elevations and/or bleeding events have been reported in some patients taking warfarin.

4.6 Fertility, pregnancy and lactation

Women of childbearing potential

Women of childbearing potential must be advised not to get pregnant during therapy.

Pregnancy

There are no data from the use of Gefitinib in pregnant women. Studies in animals have shown reproductive toxicity. The potential risk for humans is unknown. Genessa should not be used during pregnancy unless clearly necessary.

Breast-feeding

It is not known whether Gefitinib is secreted in human milk. Gefitinib and metabolites of Gefitinib accumulated in milk of lactating rats. Gefitinib is contraindicated during breast-feeding and therefore breast-feeding must be discontinued while receiving Gefitinib therapy.

4.7 Effects on Ability to Drive and Use Machines

During treatment with gefitinib, asthenia has been reported. Therefore, patients who experience this symptom should be cautious when driving or using machines.

4.8 Undesirable Effects

Adverse reactions have been assigned to the frequency categories in Table 1 where possible based on the incidence of comparable adverse event reports. Frequencies of occurrence of undesirable effects are defined as: very common , common ,uncommon, rare, very rare and not known.

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Table 1 Adverse reactions

Adverse drug reactions by system organ class and frequency		
Metabolism and nutrition disorders	Very Common	Anorexia mild or moderate (CTC grade 1 or 2)
Eye disorders	Common	Conjunctivitis, blepharitis, and dry eye*, mainly mild (CTC grade 1)
	Uncommon	Corneal erosion, reversible and sometimes in association with aberrant eyelash growth Keratitis
Vascular disorders	Common	Hemorrhage, such as epistaxis and hematuria
Respiratory, thoracic and mediastinal disorders	Common	Interstitial lung disease often severe (CTC grade 3 or 4). Cases with fatal outcomes have been reported

Gastrointestinal disorders	Very Common	Diarrhea, mainly mild or moderate (CTC grade 1 or 2) Vomiting, mainly mild or moderate (CTC grade 1 or 2) Nausea, mainly mild (CTC grade 1) Stomatitis, predominantly mild (CTC grade 1)
	Common	Dehydration, secondary to diarrhea, nausea, vomiting or anorexia Dry mouth*, predominantly mild (CTC grade 1)
	Uncommon	Pancreatitis Gastrointestinal perforation
Hepatobiliary disorders	Very Common	Elevations in alanine aminotransferase, mainly mild to moderate
	Common	Elevations in aspartate aminotransferase, mainly mild to moderate Elevations in total bilirubin, mainly mild to moderate
	Uncommon	Hepatitis**
Skin and subcutaneous tissue disorders	Very Common	Skin reactions, mainly a mild or moderate (CTC grade 1 or 2) pustular rash, sometimes itchy with dry skin, including skin fissures, on an erythematous base
	Common	Nail disorder Alopecia Allergic reactions, including angioedema and urticaria
	Rare	Bullous conditions including Toxic epidermal necrolysis, Stevens Johnson syndrome and erythema multiforme Cutaneous vasculitis
Renal and urinary disorders	Common	Asymptomatic laboratory elevations in blood creatinine Proteinuria Cystitis
	Rare	Hemorrhagic cystitis
General disorders and administration site conditions	Very Common	Asthenia, predominantly mild (CTC grade 1)
	Common	Pyrexia

The frequencies of adverse drug reactions relating to abnormal laboratory values are based on patients with a change from baseline of 2 or more CTC grades in the relevant laboratory parameters.

* This adverse reaction can occur in association with other dry conditions (mainly skin reactions) seen with Gefitinib.

** This includes isolated reports of hepatic failure which in some cases led to fatal outcomes.

4.9 Overdose

There is no specific treatment in the event of overdose of Gefitinib. Adverse reactions associated with overdose should be treated symptomatically; in particular severe diarrhea should be managed as clinically indicated. In a clinical study, a limited number of patients were treated with daily doses of up to 1000 mg. An increase of frequency and severity of some adverse reactions was observed, mainly diarrhea and skin rash. In one study a limited number of patients were treated weekly with doses from 1500 mg to 3500 mg. In this study Gefitinib exposure did not increase with increasing dose, adverse events were mostly mild to moderate in severity, and were consistent with the known safety profile of Gefitinib.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic Properties

Gefitinib is a selective inhibitor of the epidermal growth factor receptor (EGFR) tyrosine kinase, commonly expressed in solid human tumors of epithelial origin. Inhibition of EGFR tyrosine kinase activity inhibits tumor growth, metastasis and angiogenesis and increases tumor cell apoptosis.

Patients that have never smoked, have adenocarcinoma histology, are female gender or are of Asian ethnicity, are more like to benefit from treatment with Gefitinib. These clinical characteristics are also associated with a higher rate of EGFR mutation positive tumors.

Resistance
Most NSCLC tumors with sensitizing EGFR kinase mutations eventually develop resistance to Gefitinib treatment with a median time to disease progression of 1 year. In about 60% of cases, resistance is associated with a secondary T790M mutation for which T790M targeted EGFR TKIs may be considered as a next line treatment option. Other potential mechanisms of resistance have been reported following treatment with EGFR signal blocking agents including bypass signaling such as HER2 and MET gene amplification and PIK3CA mutations. Phenotypic switch to small cell lung cancer has also been reported in 5-10% of cases.

5.2 Pharmacokinetic Properties

Following intravenous administration, Gefitinib is rapidly cleared, extensively distributed and has a mean elimination half-life of 48 hours. Following oral dosing in cancer patients, absorption is moderately slow and the mean terminal half-life is 41 hours. Administration of Gefitinib once daily results in 2 to 8-fold accumulation with steady state exposures achieved after 7 to 10 doses. At steady state, circulating plasma concentrations are typically maintained within a 2 to 3-fold range over the 24-hour dosing interval.

Absorption:
Following oral administration of Gefitinib, peak plasma concentrations of Gefitinib typically occur at 3 to 7 hours after dosing. Mean absolute bioavailability is 59% in cancer patients. Exposure to Gefitinib is not significantly altered by food. In a trial in healthy volunteers where gastric pH was maintained above pH 5, Gefitinib exposure was reduced by 47%.

Distribution:
Mean volume of distribution at steady state of Gefitinib is 1400 L indicating extensive distribution into tissue. Plasma protein binding is approximately 90%. Gefitinib binds to serum albumin and α1-acid glycoprotein.

Metabolism:
In vitro data indicate that CYP3A4 is the major P450 isozyme involved in the oxidative

metabolism of Gefitinib.

In vitro studies have shown that Gefitinib has limited potential to inhibit CYP2D6. In a clinical trial in patients, Gefitinib was co-administered with metoprolol (a CYP2D6 substrate). This resulted in a small (35%) increase in exposure to metoprolol, which is not considered to be clinically relevant.

Gefitinib shows no enzyme induction effects in animal studies and no significant inhibition (*in vitro*) of any other cytochrome P450 enzyme.

Three sites of biotransformation have been identified in the metabolism of Gefitinib: metabolism of the N-propylmorpholino-group, demethylation of the methoxy- substituent on the quinazoline and oxidative defluorination of the halogenated phenyl group. Five metabolites have been fully identified in fecal extracts and the major component was O-desmethyl gefitinib, although this only accounted for 14% of the dose.

In human plasma 8 metabolites were fully identified. The major metabolite identified was O-desmethyl gefitinib, which was 14-fold less potent than Gefitinib at inhibiting EGFR stimulated cell growth and had no inhibitory effect on tumor cell growth in mice. It is therefore considered unlikely that it contributes to the clinical activity of Gefitinib.

The production of O-desmethyl gefitinib has been shown, in vitro, to be via CYP2D6. The role of CYP2D6 in the metabolic clearance of Gefitinib has been evaluated in a clinical trial in healthy volunteers genotyped for CYP2D6 status. In poor metabolizers, no measurable levels of O-desmethyl gefitinib were produced. The range of Gefitinib exposures achieved in both the extensive and the poor metabolizer groups were wide and overlapping but the mean exposure to Gefitinib was 2-fold higher in the poor metabolizer group. The higher average exposures that could be achieved by individuals with no active CYP2D6 may be clinically relevant since adverse experiences are related to dose and exposure.

Elimination:
Gefitinib total plasma clearance is approximately 500 mL/min. Excretion is predominantly via the feces with renal elimination of drug and metabolites accounting for less than 4% of the administered dose.

Special populations:
In population based data analyses in cancer patients, no relationships were identified between predicted steady state trough concentration and patient age, body weight, gender, ethnicity or creatinine clearance.

In a study of single dose Gefitinib 250mg in patients with mild, moderate or severe hepatic impairment due to cirrhosis (according to Child-Pugh classification), there was an increase in exposure in all groups compared with healthy controls. An average 3.1-fold increase in exposure to Gefitinib in patients with moderate and severe hepatic impairment was observed. None of the patients had cancer, all had cirrhosis and some had hepatitis. This increase in exposure may be of clinical relevance since adverse experiences are related to dose and exposure to Gefitinib.

Gefitinib has been evaluated in a clinical trial conducted in 41 patients with solid tumors and normal hepatic function or, moderate or severe hepatic dysfunction due to liver metastases. It was shown that following daily dosing of 250 mg Gefitinib, time to steady state, total plasma clearance and steady state exposure (C_{maxSS}, AUC_{24SS}) were similar for the groups with normal and moderately impaired hepatic function. Data from 4 patients with severe hepatic dysfunction due to liver metastases suggested that steady state exposures in these patients are also similar to those in patients with normal hepatic function.

6 PHARMACEUTICAL PARTICULARS

6.1 List of Excipients

Tablet core:

Lactose Monohydrate, Microcrystalline Cellulose, Crospovidone, Povidone, Sodium Lauryl Sulfate, Magnesium Stearate

Film coat:

Opadry II Complete Film Coating System 85F565317 Brown

6.2 Incompatibilities

Not applicable

6.3 Shelf Life

Please refer to the expiry date on the product labels.

6.4 Special Precautions for Storage

Do not store above 30°C, keep in the original package in order to protect from moisture.

6.5 Nature and Content of Container

Alu/Alu blisters of 1x10’s, 3x10’s, 6x10’s, 10x10’s, 50x10’s and 100x10’s tablets packed in the outer carton. Not all pack size will be marketed.

6.6 Special precautions for disposal and other handling

No special requirements.

7 MANUFACTURER

Manufactured by:
Novugen Oncology Sdn. Bhd.
No. 47, Jalan Lengkuk Teknologi 2,
Taman Teknologi Enstek Fasa 1,
71760 Bandar Baru Enstek,
Negeri Sembilan, Malaysia.

Product Registration Holder:

Novugen Oncology Sdn. Bhd.
No. 3, Jalan Jururancang U1/21,
Hicom Glenmarie Industrial Park,
40150 Shah Alam, Selangor, Malaysia.

8 DATE OF REVISION

19/05/2023

Package insert is drafted in Times New Roman with font size of 12, subject to change based on any other legible font and font size.